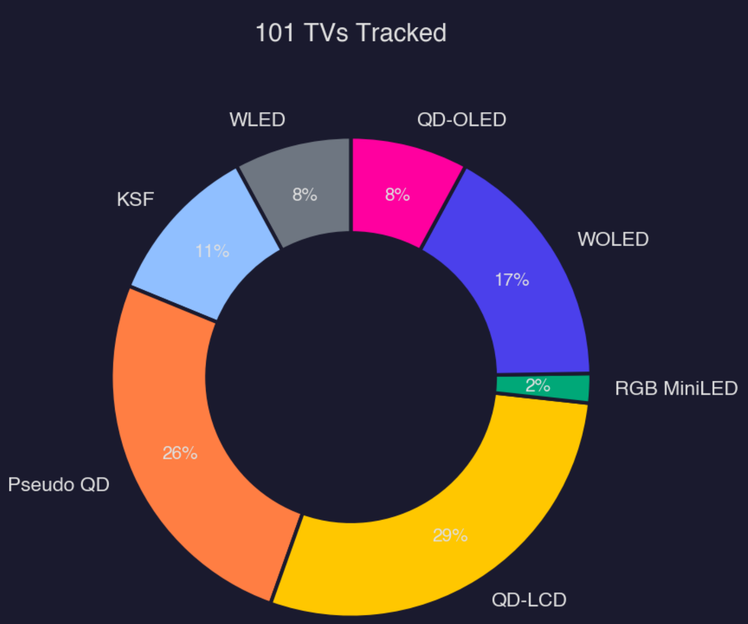
Display Technology Intelligence Report

June 2026

101 4K TVs tracked | 86 with pricing | 6 technologies | 3 8K excluded



1. Display Technology Overview

Section 1: Display Technology Overview -- June 2026

1.1 New & Removed Devices

The June 2026 database stands at 101 TVs (excluding 3 8K models), with five new entries added this month and zero removals -- a clean, additive update. All five arrivals share a single panel architecture: QD-LCD. The TCL QM7L, Hisense UR9SG, and Samsung R95H arrive as the stronger performers in the cohort, posting mixed-usage scores of 8.2, 8.4, and 8.3 respectively. The Amazon Ember Artline 2026 (6.2) and Samsung M70H (5.6) land at the lower end of the QD-LCD range, reinforcing that the architecture spans a wide capability spectrum depending on implementation quality.

The absence of removals signals manufacturer lineup stability heading into mid-year -- no legacy models were quietly discontinued or pulled from tracking. The QD-LCD-only intake is itself a data point: it reflects the continued investment manufacturers are making in quantum dot LCD as the mainstream performance tier, rather than pushing new OLED or emerging RGB MiniLED entrants at this stage of the cycle.

The sustained QD-LCD momentum in new arrivals positions that technology for further share gains in the tracked database, which will sharpen competitive pressure on the Pseudo QD and KSF tiers sitting just below it in performance.

1.2 Performance Snapshot by Technology

QD-OLED leads the database across all three use-case scores and it isn't close at the top. Its mean mixed-usage score of 8.7, home theater score of 8.9, and gaming score of 9.0 make it the only technology to clear 9.0 in any category. Its floor scores are also the highest of any technology -- a minimum mixed-usage of 8.2 means even the weakest QD-OLED in the database outperforms the average QD-LCD set. WOLED runs second across the board, posting means of 8.4 (mixed), 8.6 (home theater), and 8.7 (gaming), with a ceiling of 9.2 in both mixed and home theater -- matching QD-OLED's top end but trailing in average consistency.

RGB MiniLED earns attention despite representing only 2 models in the database. Its mean mixed-usage of 8.4 ties WOLED, and its home theater mean of 8.5 edges WOLED's 8.6 to within rounding distance -- a striking result for a nascent category. QD-LCD sits as the clear LCD-architecture leader with a mean mixed-usage of 7.9 and a maximum of 8.6, but it trails OLED-class technologies by roughly 0.8 points on average. The gap between QD-LCD and the Pseudo QD tier (6.6 mean mixed) is 1.3 points -- larger than the gap between Pseudo QD and WLED (5.2), suggesting QD implementation quality creates the sharpest performance discontinuity in the LCD stack. WLED sits at the bottom with a mean gaming score of just 5.1 and a mixed-usage ceiling of 5.6.

The 0.2-point QD-OLED lead over WOLED in gaming (9.0 vs. 8.7) is the most commercially meaningful gap in the data -- gaming remains the highest-frequency premium use case, and QD-OLED's structural advantage there will continue to drive its positioning in enthusiast retail channels.

1.3 FWHM Spectral Analysis

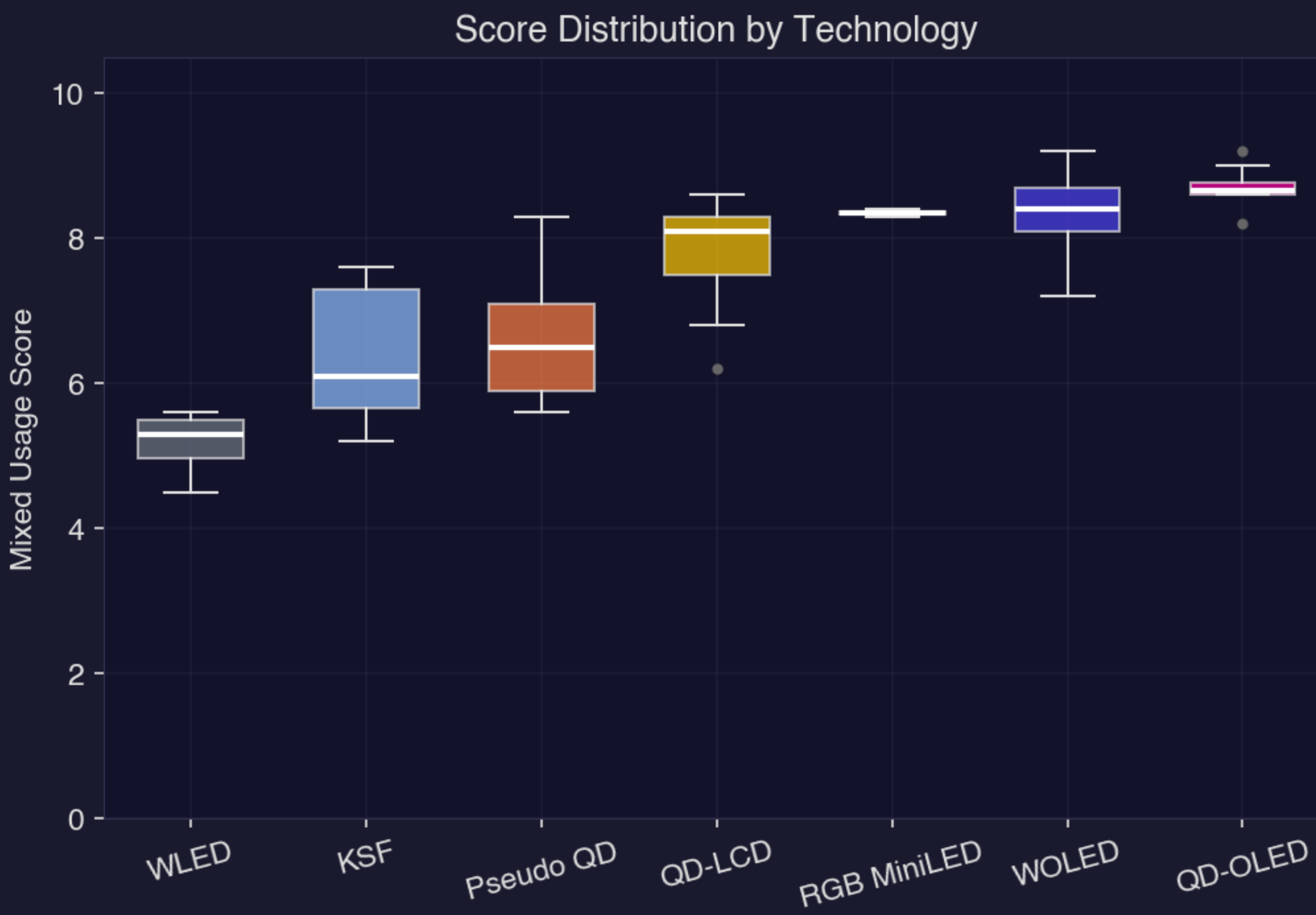
Full-width at half-maximum (FWHM) measurements quantify how spectrally pure each technology's light emission is -- narrower values mean more saturated, precise color. QD-LCD and RGB MiniLED are the LCD-architecture leaders here. QD-LCD posts green and red FWHM means of 29.3 nm and 28.8 nm respectively, while RGB MiniLED is even tighter at 26.9 nm green and 16.5 nm red -- the narrowest red peak in the entire database by a wide margin. That sub-17 nm red is what enables RGB MiniLED's color volume performance and helps explain its competitive home theater scores despite limited market representation.

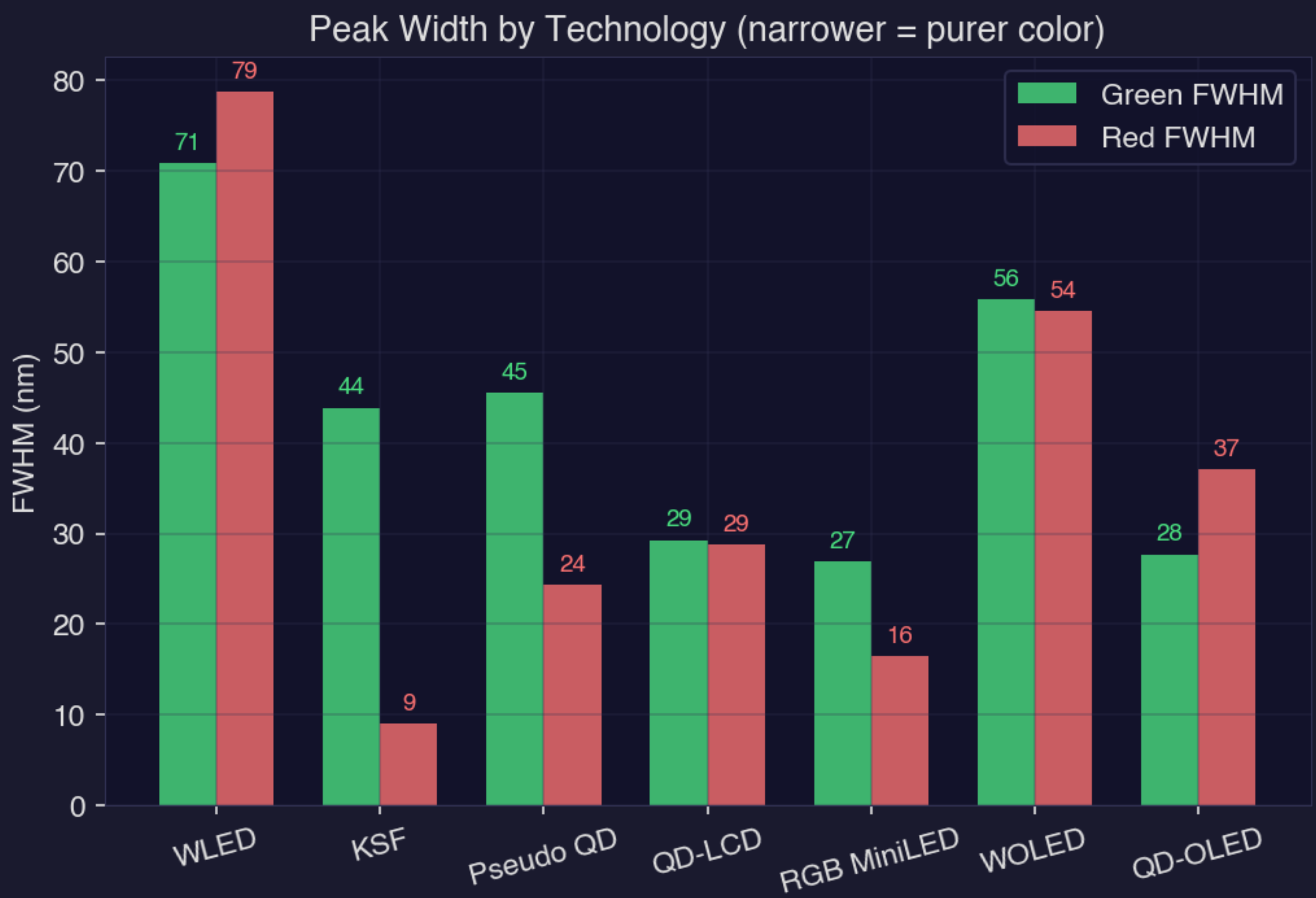
KSF is the spectral surprise: its red FWHM of 9.1 nm is extraordinarily narrow, the tightest red emission in the database, a consequence of its phosphor-based red emitter design. Its green at 43.9 nm is wider, but the red precision gives KSF sets a specific advantage in saturated red reproduction that its middling overall scores (mean mixed-usage 6.4) don't fully capture. WLED sits at the opposite extreme -- green FWHM of 70.9 nm and red of 78.7 nm -- broad, overlapping emission profiles that directly explain the color accuracy limitations reflected in its low scores. WOLED also shows wide green emission at 83.7 nm, a known structural characteristic of its white-emitter-plus-color-filter architecture, yet its self-emissive contrast properties allow it to overcome this spectral disadvantage at the system level.

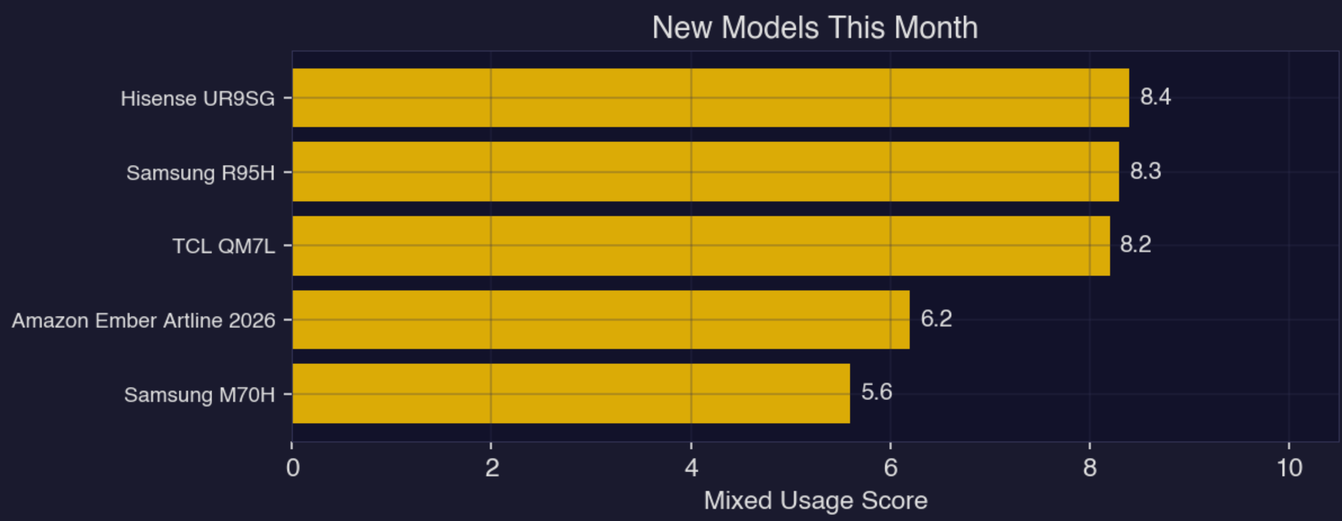
QD-OLED's green FWHM of 27.7 nm is the narrowest among self-emissive technologies and rivals QD-LCD's purity -- combined with its already-leading performance scores, this confirms QD-OLED as the current apex of simultaneous color precision and contrast capability. The spectral gap between QD-OLED and WOLED at the green channel (27.7 nm vs. 83.7 nm) will keep color gamut coverage as a differentiating spec in premium OLED marketing through the rest of 2026.

1.4 Model Year Comparison

The 2026 model cohort currently totals 18 TVs against 41 tracked 2025 models, so year-over-year conclusions carry some statistical weight but will sharpen as more 2026 releases enter the database in coming months. On the available data, 2026 models score 7.8 on average for mixed usage versus 7.3 for


Score Distribution by Technology
Mixed Usage Score
0
2
4
6
8
10
WLED
KSF
Pseudo QD
QD-LCD
RGB MiniLED
WOLED
QD-OLED


Peak Width by Technology (narrower = purer color)
Green FWHM
Red FWHM
FWHM (nm)
0
10
20
30
40
50
60
70
80
71
79
44
9
45
24
29
29
27
16
56
54
28
37
WLED
KSF
Pseudo QD
QD-LCD
RGB MiniLED
WOLED
QD-OLED


New Models This Month
Hisense UR9SG
Samsung R95H
TCL QM7L
Amazon Ember Artline 2026
Samsung M70H
8.4
8.3
8.2
6.2
5.6
0
2
4
6
8
10
Mixed Usage Score

2. Pricing Intelligence

Section 2: Pricing Intelligence

2.1 Current Pricing Landscape

The 86-model dataset tracked through June 2026 spans a price-per-square-meter range from $574 (WLED) to $3,211 (RGB MiniLED) -- a 5.6x spread that underscores just how fractured the TV market remains across technology tiers. At the budget end, WLED holds the floor with a median set price of $430, while KSF and Pseudo QD cluster just above it at $980 and $771 respectively, forming a distinct mid-market band that competes heavily on value rather than specification supremacy.

The premium tier tells a different story. QD-OLED commands a median price of $2,179 and WOLED sits at $2,012, making them the dominant players in the $2,000+ segment. RGB MiniLED, however, is the true outlier -- its $2,500 median and $3,211 per-square-meter cost make it the single most expensive technology in the dataset, surpassing even QD-OLED on an area-normalized basis by $805/m2. That premium reflects the nascent manufacturing economics of full-color-channel MiniLED arrays, where panel yields and component costs haven't yet benefited from scale.

The mid-to-premium gap is particularly stark: QD-LCD at $1,484 median sits in relative isolation between the sub-$1,000 LCD tiers and the $2,000+ OLED/MiniLED cluster, positioning it as the entry point into serious home cinema performance for consumers unwilling to cross the four-figure threshold into OLED territory.

2.2 Price Trends

With 19 weeks of pricing history in the dataset, directional signals are clear: every technology tracked declined in per-square-meter cost over the observation window, but the magnitude varies dramatically by tier. WOLED led all declines at -10.4%, falling from $2,517/m2 to $2,256/m2 -- a compression that reflects both LG Display's push to move panel volume ahead of the summer selling season and intensifying competition from QD-OLED at similar price points.

KSF dropped -6.6% (from $1,070 to $1,000/m2) and QD-OLED fell -6.8% (from $2,582 to $2,406/m2), both registering meaningful deflation that suggests manufacturers are actively repricing to stimulate demand rather than holding margin. Pseudo QD and QD-LCD moved in a tighter band, declining -3.2% and -2.9% respectively, while WLED was essentially flat at -0.6% -- unsurprising for a commodity technology with little remaining pricing flexibility.

The WOLED and QD-OLED simultaneous declines are the most consequential trend: as both technologies compress toward each other, the consumer decision between them increasingly shifts from price to panel preference, which puts greater pressure on picture quality differentiation and retail merchandising to drive conversion.

2.3 Value Analysis

On a dollars-per-performance-point basis, the Vizio Mini LED Quantum 4K is the standout of the June dataset -- a $218 KSF-architecture set delivering a mixed-usage score of 6.3 at just $35 per point, the lowest efficiency ratio across all 86 models tracked. That figure is roughly 40% cheaper per point than the next-best performer, the Samsung U7900F, which costs $300 but scores only 5.2 on mixed usage, translating to $58/point. The Vizio's performance-to-price ratio is exceptional by any measure in this dataset.

The KSF and Pseudo QD architectures dominate the value leaderboard. The TCL S551G (KSF, $370, score 5.9, $63/point) and Hisense QD6QF (Pseudo QD, $370, score 5.9, $63/point) land at identical cost efficiency -- a statistical tie that reflects how closely matched these two color architectures have become at retail. The TCL Q651G closes the top-five at $66/point with a $380 price tag and 5.8 mixed-usage score, confirming TCL's consistent execution across both KSF and Pseudo QD product lines.

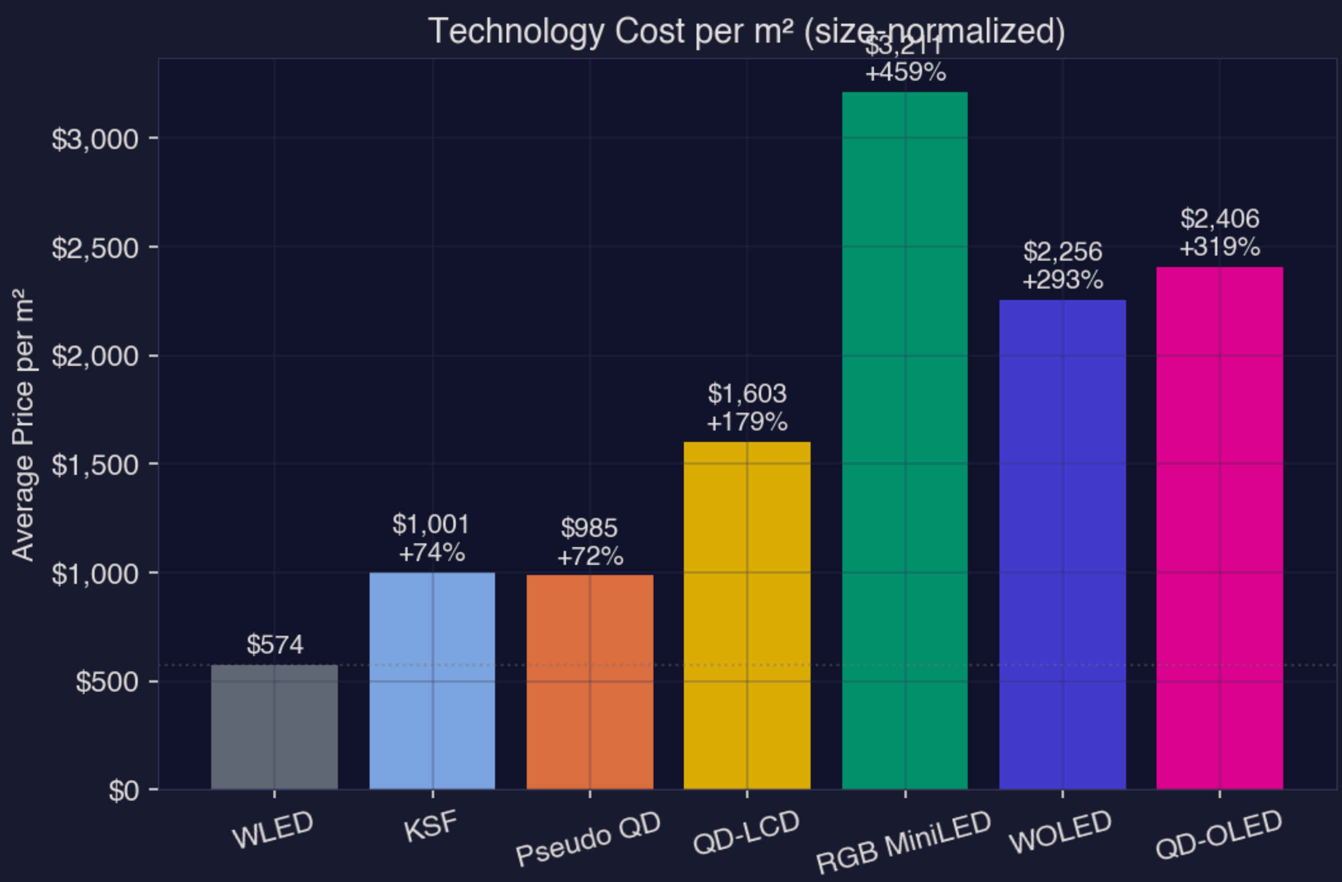
Premium technologies, by contrast, extract a steep efficiency penalty. KSF commands a 74% premium per square meter over WLED; QD-OLED sits at +319%; and RGB MiniLED reaches +459% -- a gap that the performance scores, while higher in absolute terms, do not yet fully justify for mainstream consumers comparing receipts at the point of sale.

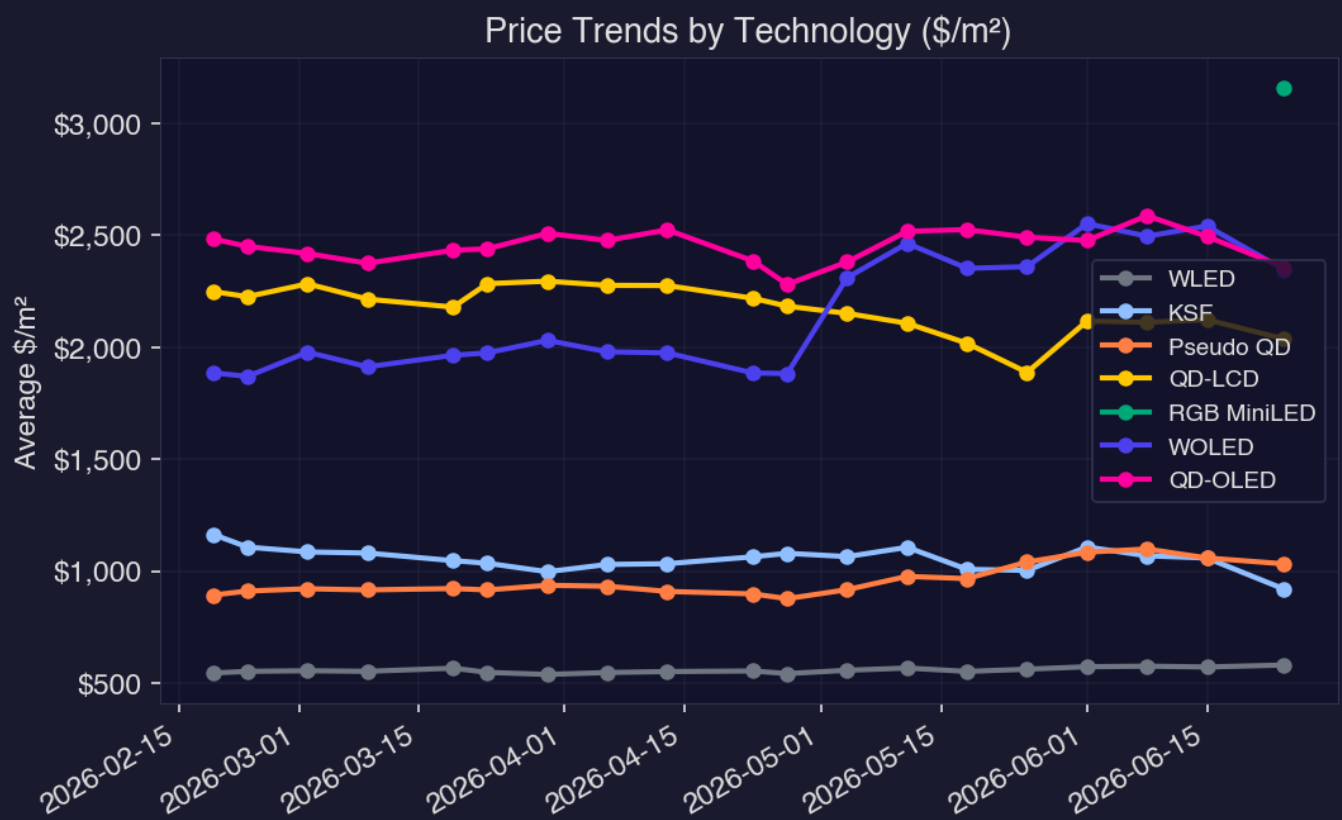
2.4 Notable Pricing Gaps

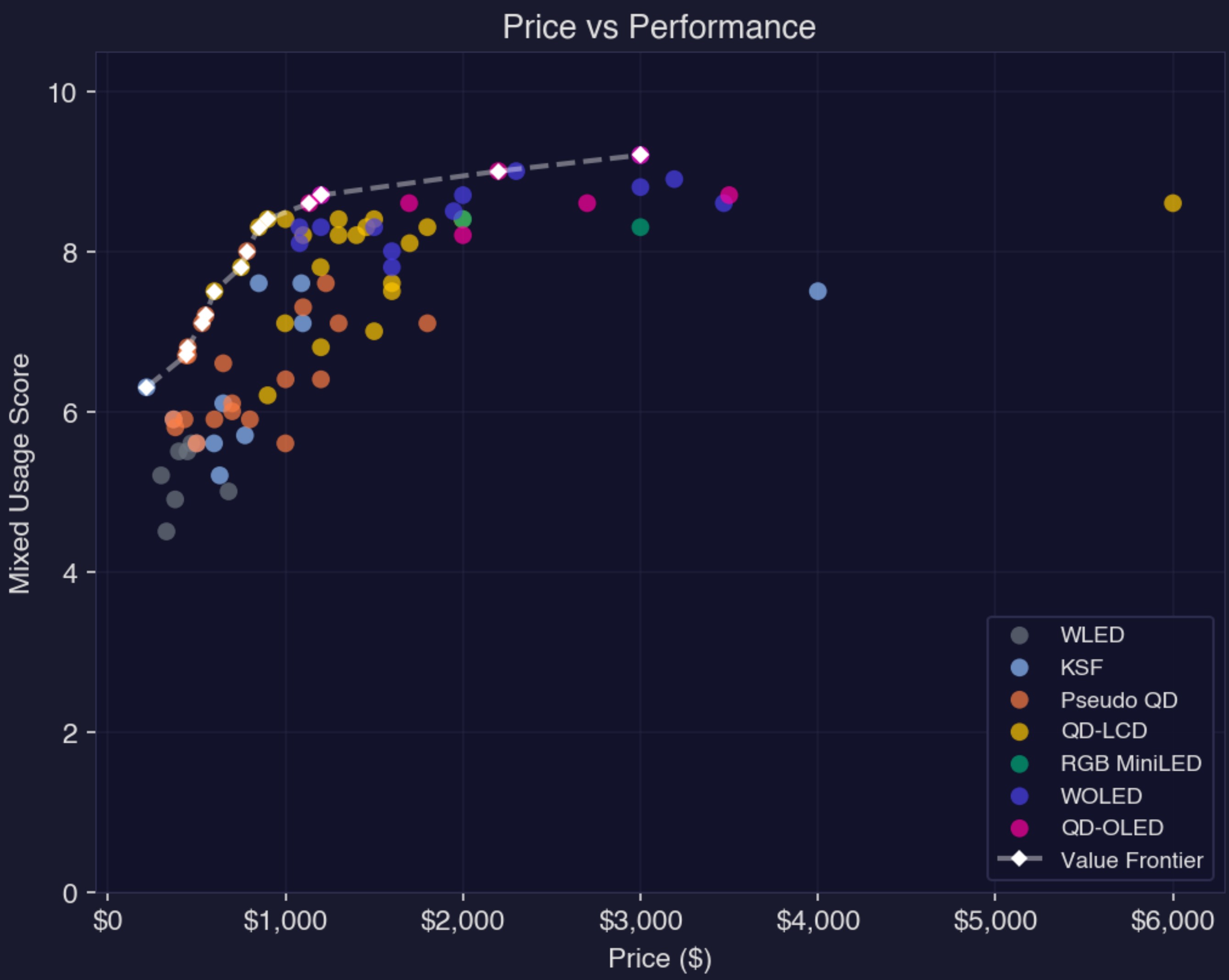
The most commercially significant pricing gap in the current market sits between Pseudo QD ($985/m2) and QD-LCD ($1,997/m2) -- a $1,012/m2 jump for what is nominally the next step up the technology ladder. That chasm creates a no-man's-land where a consumer moving beyond entry-level quantum dot performance faces an immediate doubling of per-area cost before reaching a true full-array local dimming, wide-color-gamut QD-LCD product. Retailers filling that gap with stretched QD-LCD inventory at promotional prices are the ones best positioned to convert mid-market shoppers this summer.

At the top of the stack, the $805/m2 gap between RGB MiniLED and QD-OLED represents a premium that currently flows entirely to panel novelty rather than demonstrated performance leadership in most use cases. WOLED, meanwhile, now sits only $150/m2 below QD-OLED after its -10.4% price decline -- the narrowest that gap has been in the 19-week observation window -- effectively making WOLED the more aggressive value proposition in the OLED segment for price-sensitive premium buyers.

As WOLED and QD-OLED pricing continues to converge, the retailers and brands that fail to clearly articulate the technical distinctions between the two panel types at shelf


Technology Cost per m² (size-normalized)
Average Price per m²
$0
$500
$1,000
$1,500
$2,000
$2,500
$3,000
$574
$1,001
+74%
$985
+72%
$1,603
+179%
$3,211
+459%
$2,256
+293%
$2,406
+319%
WLED
KSF
Pseudo QD
QD-LCD
RGB MiniLED
WOLED
QD-OLED


Price Trends by Technology ($/m²)
Average $/m²
$500
$1,000
$1,500
$2,000
$2,500
$3,000
2026-02-15
2026-03-01
2026-03-15
2026-04-01
2026-04-15
2026-05-01
2026-05-15
2026-06-01
2026-06-15
WLED
KSF
Pseudo QD
QD-LCD
RGB MiniLED
WOLED
QD-OLED


Price vs Performance
Mixed Usage Score
10
8
6
4
2
0
$0
$1,000
$2,000
$3,000
$4,000
$5,000
$6,000
Price ($)
WLED
KSF
Pseudo QD
QD-LCD
RGB MiniLED
WOLED
QD-OLED
Value Frontier

3. Macro & Industry Context

Section 3: Macro & Industry Context

The Display Technology Stack, June 2026

Our database of 101 televisions across 6 distinct display technologies captures a market in the middle of a structural shift -- one where the gap between marketing language and physical reality has never been wider or more consequential. The technology distribution tells the story directly: QD-LCD leads with 29 panels, followed closely by Pseudo QD at 26 -- a category that exists almost entirely as a labeling strategy, not an engineering distinction. Combined, those two buckets account for 54% of tracked panels, meaning the majority of the market is still built on LCD backplanes dressed in quantum dot branding of varying legitimacy. Meanwhile, the genuine self-emissive technologies -- WOLED at 17 units and QD-OLED at 8 -- represent premium execution at a fraction of the unit count. RGB MiniLED, at just 2 panels, signals an emerging segment that has not yet scaled.

Samsung's dominance at 31 panels and Hisense's 18 reveal two very different growth vectors. Samsung's count spans QD-LCD and QD-OLED, leveraging its own display manufacturing to own multiple tiers simultaneously. Hisense, along with TCL at 15 panels, represents the Chinese manufacturer push into every technology bracket -- including aggressive miniLED backlighting in QD-LCD panels -- compressing the premium tier from below. LG's 14 panels skew heavily toward WOLED and QD-OLED, a deliberate bet on self-emissive as the defensible moat. The brand distribution across our database is not random; it mirrors exactly which companies are investing in panel fabrication versus which are assembling commodity components into branded boxes.

Technology Differentiation: What the Specs Actually Mean

The practical performance gap between these technologies is large and measurable. QD-OLED panels achieve peak brightness figures -- typically in the 1,500 - 2,000 nit range for small-window highlights -- while delivering the pixel-level black floors that only self-emissive architectures can produce, yielding contrast ratios that LCD-based panels cannot match structurally. WOLED operates at somewhat lower peak brightness but covers the full lineup from 55 to 97 inches, a size scalability that QD-OLED has not yet matched at equivalent cost. The critical distinction between the two OLED variants is color volume: QD-OLED's quantum dot conversion layer pushes wider color gamut at high brightness, where WOLED's white subpixel dilutes saturation as luminance climbs.

The gap between true QD-LCD and Pseudo QD is where consumer confusion is most commercially exploited. Genuine quantum dot LCD panels use cadmium-free InP or similar nanocrystal films to narrow the red and green emission spectra, producing DCI-P3 coverage typically above 90%. Pseudo QD panels -- tracked separately in our database as 26 units -- substitute KSF red phosphor and market the result under quantum dot branding, delivering meaningfully narrower color gamut. WLED, at 8 panels, is standard phosphor LCD with no spectral engineering at all. These three technologies represent three distinct performance tiers that retail shelves routinely flatten into a single "LED TV" category, creating systematic information asymmetry that benefits volume-oriented manufacturers.

Where the Industry Is Heading

Three technology vectors are converging toward the next platform inflection. Tandem OLED -- stacking two emissive layers to dramatically increase brightness without accelerating burn-in -- is already appearing in tablet applications and will reach television panel sizes within the 2026 - 2027 production cycle. This directly addresses WOLED's brightness ceiling and would compress the QD-OLED brightness advantage considerably. MicroLED, currently represented by only 2 RGB MiniLED panels in our database, remains constrained by mass transfer yield economics, but per-unit manufacturing costs have fallen roughly 30 - 40% over the past three years; the technology is moving from installation art to aspirational consumer product on a credible timeline. Perovskite quantum dots represent the most disruptive long-term variable -- their theoretical color purity exceeds conventional InP quantum dots substantially, and stability concerns that plagued early research have narrowed considerably, with pilot production lines already operating in Asia.

The "so what" for this team is structural: the Pseudo QD category's 26-panel presence in our database is a commercial opportunity masquerading as a technical category, and the brands willing to draw a hard line between genuine QD-LCD and phosphor substitutes will own the premium LCD narrative as self-emissive technologies absorb the top of the market.

4. Year-over-Year Technology Trends

Section 4: Year-over-Year Technology Trends

4.1 Score Trajectories: Who Improved, Who Didn't

The clearest winner in mixed-usage performance between 2025 and 2026 is WOLED, which climbed +0.4 points to reach an average score of 8.8 -- the highest of any display technology tracked this cycle. That gain is meaningful at the top of the scale, where each decimal point reflects hard engineering tradeoffs in brightness, color volume, and motion handling. QD-LCD also advanced, moving from 7.7 to 7.9, a modest but consistent improvement that signals steady panel and backlight refinement from Samsung and its supply chain partners.

The more complicated story belongs to KSF and Pseudo QD. KSF -- present in just 3 models this year versus 2 last year -- saw its average score drop sharply from 7.3 to 6.0, a -1.3 point decline that suggests the newer additions to the category are lower-tier implementations rather than flagship-grade panels. Pseudo QD reversed course in the opposite direction, improving +0.5 points to 7.1 despite a dramatic contraction in model count from 12 to just 3, implying the weaker performers in that tier exited the market and only the strongest specimens remain in the dataset.

The net picture from the temporal_scores chart is a modest bifurcation: WOLED and QD-LCD are pulling slightly ahead, while KSF's score dilution raises questions about where that technology is being deployed and by whom.

4.2 Pricing Dynamics: Compression, Expansion, and the Outlier

The pricing shifts between 2025 and 2026 are, in aggregate, more dramatic than the score changes. KSF posted a -57.4% collapse in price per square meter, falling from $1,665/m2 to just $709/m2 -- the most aggressive commoditization of any technology in the dataset. That kind of deflation typically signals either panel oversupply, loss of premium positioning, or both. Given that KSF's average score also dropped, the technology appears to be migrating down-market rather than expanding into new premium tiers.

At the other extreme, Pseudo QD surged +177.4% in price per square meter, from $710/m2 to $1,971/m2. That figure demands careful interpretation: with the model count collapsing from 12 to 3, the remaining Pseudo QD sets are almost certainly the top-of-segment configurations, pulling the average price dramatically upward. This isn't broad Pseudo QD inflation -- it's survivor bias in a thinning category. WOLED and QD-LCD both saw measured price increases of +25.8% and +32.0% respectively, with WOLED now sitting at $2,837/m2, reinforcing its position as the uncontested premium tier.

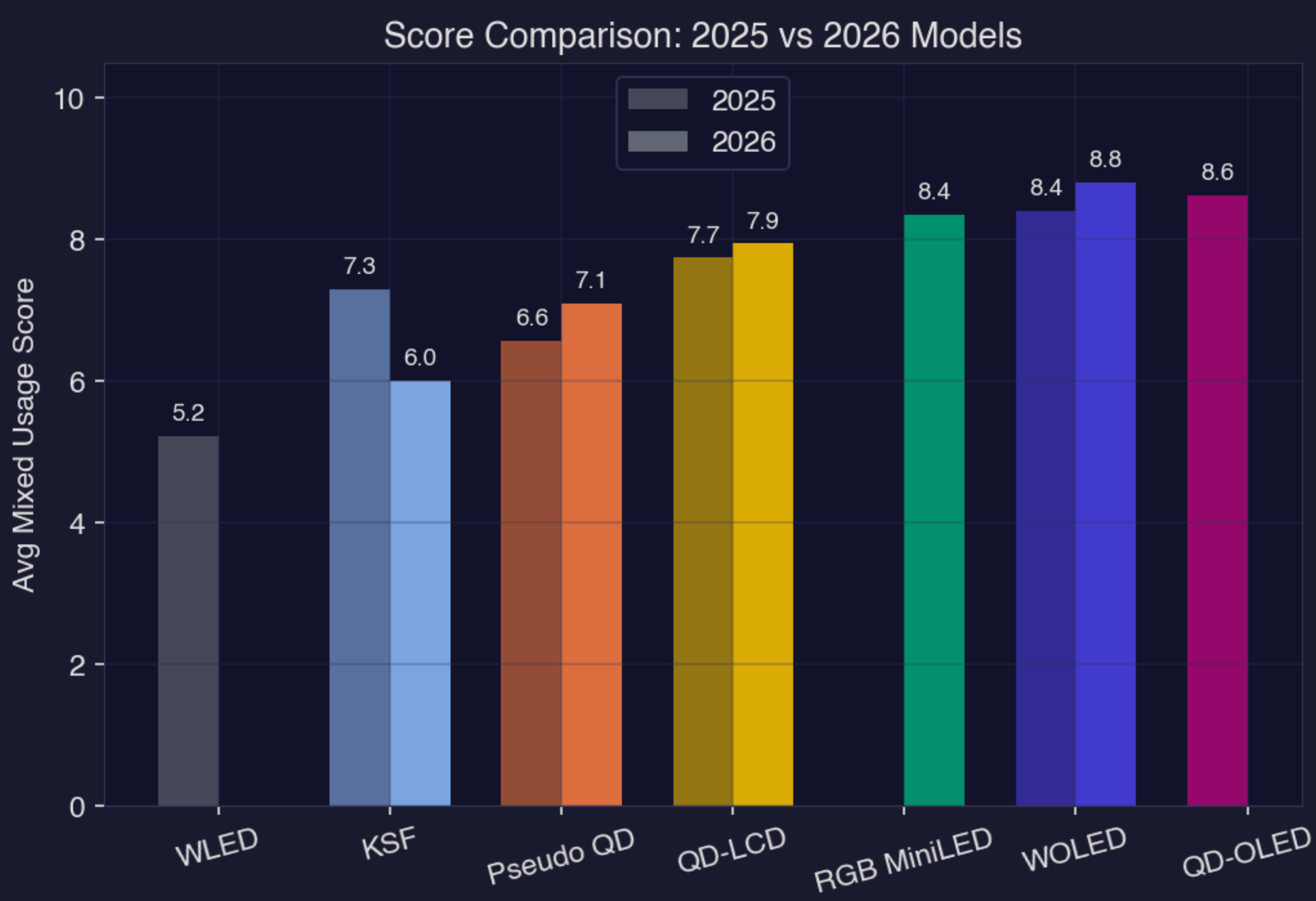
The temporal_pricing chart makes the divergence visceral: KSF and Pseudo QD are moving in opposite pricing directions for structurally different reasons, while WOLED and QD-LCD are rising in tandem -- a coherent premium escalation story.

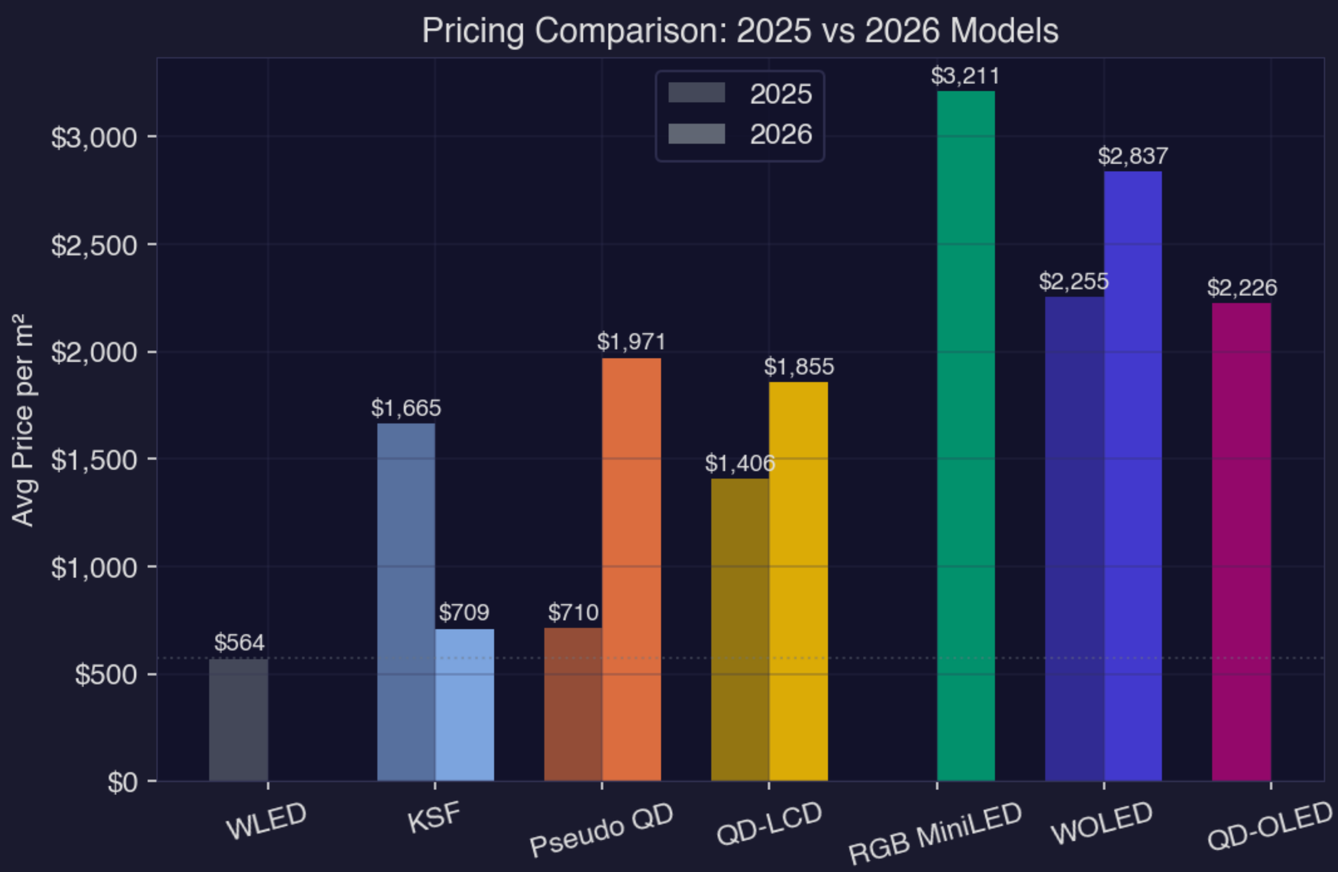
4.3 Competitive Positioning: The QD-vs-Non-QD Gap

The gap between quantum dot and non-quantum dot technologies is not narrowing -- it's calcifying. QD-LCD's 7.9 average score sits 1.9 points above KSF's 6.0, a spread that widened by 1.5 points year over year as KSF degraded. Even against Pseudo QD's improved 7.1, QD-LCD maintains a 0.8-point advantage while also commanding a higher and rising price per square meter. The data does not support any near-term convergence narrative between the tiers.

WOLED's 8.8 score at $2,837/m2 occupies a separate competitive plane entirely -- it competes less with QD-LCD on specs and more on form factor, panel uniformity, and brand positioning. The real battle remains QD-LCD versus the mid-tier alternatives, and that contest is increasingly one-sided. Manufacturers still fielding KSF or entry-grade Pseudo QD panels are defending share on price alone, not performance.

As panel yields improve and mini-LED backlight costs continue to fall, QD-LCD's price-performance ratio will tighten further -- making the case for non-QD LCD increasingly difficult to sustain past 2027.




Pricing Comparison: 2025 vs 2026 Models
2025
2026
Avg Price per m²
$3,000
$2,500
$2,000
$1,500
$1,000
$500
$0
$564
$1,665
$709
$710
$1,971
$1,406
$1,855
$3,211
$2,255
$2,837
$2,226
WLED
KSF
Pseudo QD
QD-LCD
RGB MiniLED
WOLED
QD-OLED